WIRO · XR1710G · Beginner-Friendly Flashing Guide

XR1710G Flashing Guide

Installing the Wiro iStoreOS firmware, start to finish

From opening the case and wiring up TTL to flashing the U-Boot bootloader and installing the Wiro iStoreOS firmware — every step with screenshots.

Device: XR1710G (W1700K / W1701K)·Firmware: Wiro iStoreOS Community
Audience: first-time flashers·Estimated time: 30-60 minutes

Before You Start

> **Flashing carries risks — proceed carefully.** You are responsible for any damage caused by flashing. Note that the "double clean" wipes **all** configuration and data on the router and cannot be undone.

This tutorial walks you through three things:

1. Flash a new bootloader (U-Boot) onto the router;

2. Enter Wiro recovery mode through the bootloader;

3. Install the Wiro iStoreOS firmware. Done.

What you need

Item	Notes
XR1710G router	The device to flash; open the case with a T10 screwdriver
USB-to-TTL adapter	CH340 / CH343 based ones all work (a few dollars online)
Dupont wires	3 female-to-female, to reach the TTL pins on the mainboard
Windows PC	needs one USB port and one Ethernet port
Ethernet cable	into the router's yellow 1G port
T10 screwdriver	for the case screws

Downloads

Firmware and U-Boot file (either source works):
GitHub: github.com/orangeyoo/XR1710G-OpenWrt-iStoreOS-Community
123 Cloud: 1817608703.share.123pan.cn/123pan/F32Tjv-PaNUv

Serial terminal MobaXterm (free edition is fine): mobaxterm.mobatek.net

TFTP tool 3CDaemon: included in the cloud-drive package, no installation needed.

Two files from the package are used in this tutorial — download them first:

`xr1710g-chainloader-slot.bin` — the U-Boot bootloader (chapter 4)

`istoreos-xr1710g-community-…sysupgrade.itb` — the firmware (chapter 5)

> Stuck? **QQ group: 1061612207** · Paid remote flashing: **QQ 50848**.

Chapter 1 Open the case and wire up the TTL serial adapter

A TTL serial adapter is a walkie-talkie between your PC and the router's mainboard: the PC uses it to send commands to the router one line at a time. This chapter connects that channel.

1.1 Open the case and find the TTL pins

Remove the bottom screws with the T10 screwdriver and open the case. On the edge of the mainboard you will find a row of 4–5 metal pins labeled GND, TX and RX — that is the TTL interface. The three pins are marked in the picture below.

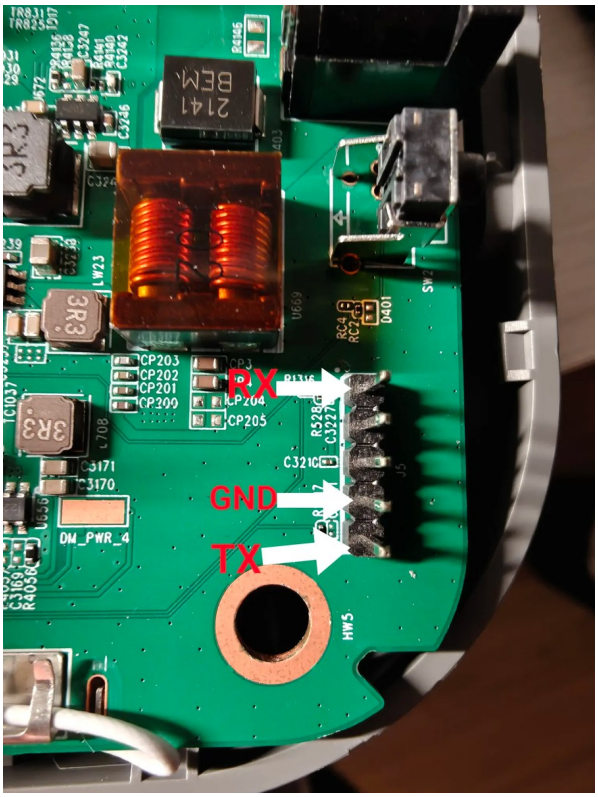


Figure 1-1 TTL pins on the mainboard (RX / GND / TX marked)

1.2 Wire the TTL adapter to the mainboard

Connect the wires per the table below. RX and TX must be **crossed over**; leave VCC unconnected to avoid power conflicts. Make sure the voltage jumper on the adapter is set to 3.3V.

Adapter pin	Mainboard pin	Notes
GND	GND	Ground — required
RX	TX	signal line, crossed
TX	RX	signal line, crossed
VCC	leave open	avoids power conflicts

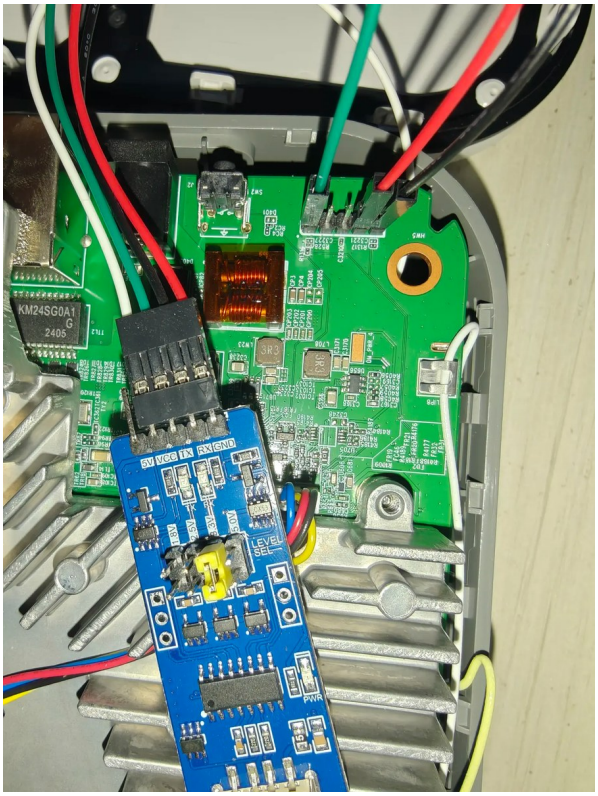


Figure 1-2 All wired up (the blue board is the USB-to-TTL adapter)

1.3 Plug into the PC and confirm the COM port

Plug the adapter's USB end into the PC. Right-click This PC → Manage → Device Manager → Ports (COM & LPT): if USB-SERIAL CH343 (COM4) shows up, the driver is fine. Remember the COM number in brackets — you will need it.

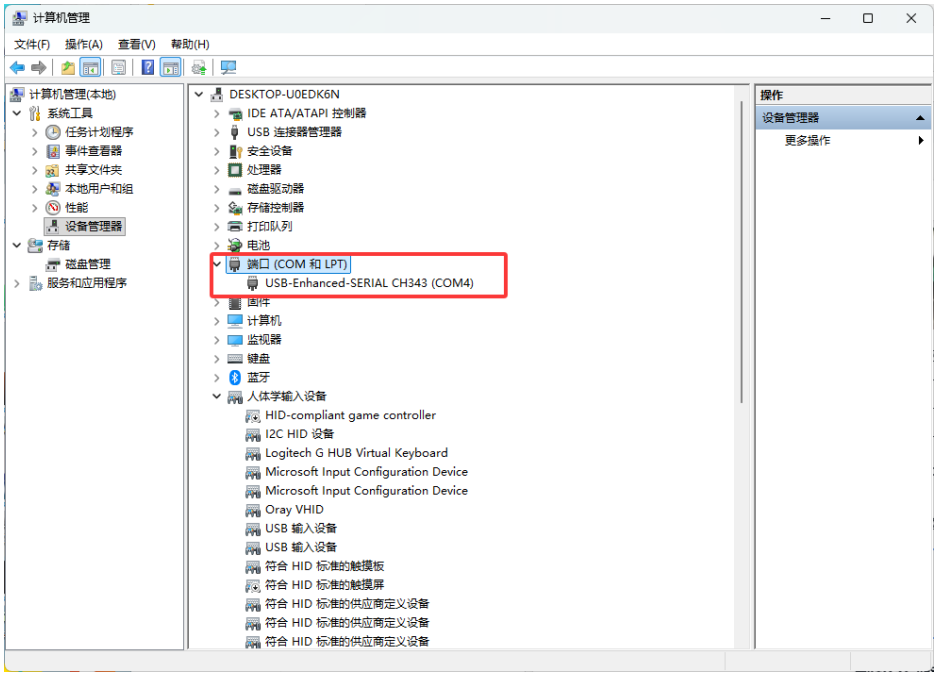


Figure 1-3 Confirming the serial port in Device Manager (COM4 in this example)

1.4 Open a serial session in MobaXterm

In MobaXterm click Session → Serial: pick your COM port (COM4 in this example), set Speed to 115200, click OK and open the session.

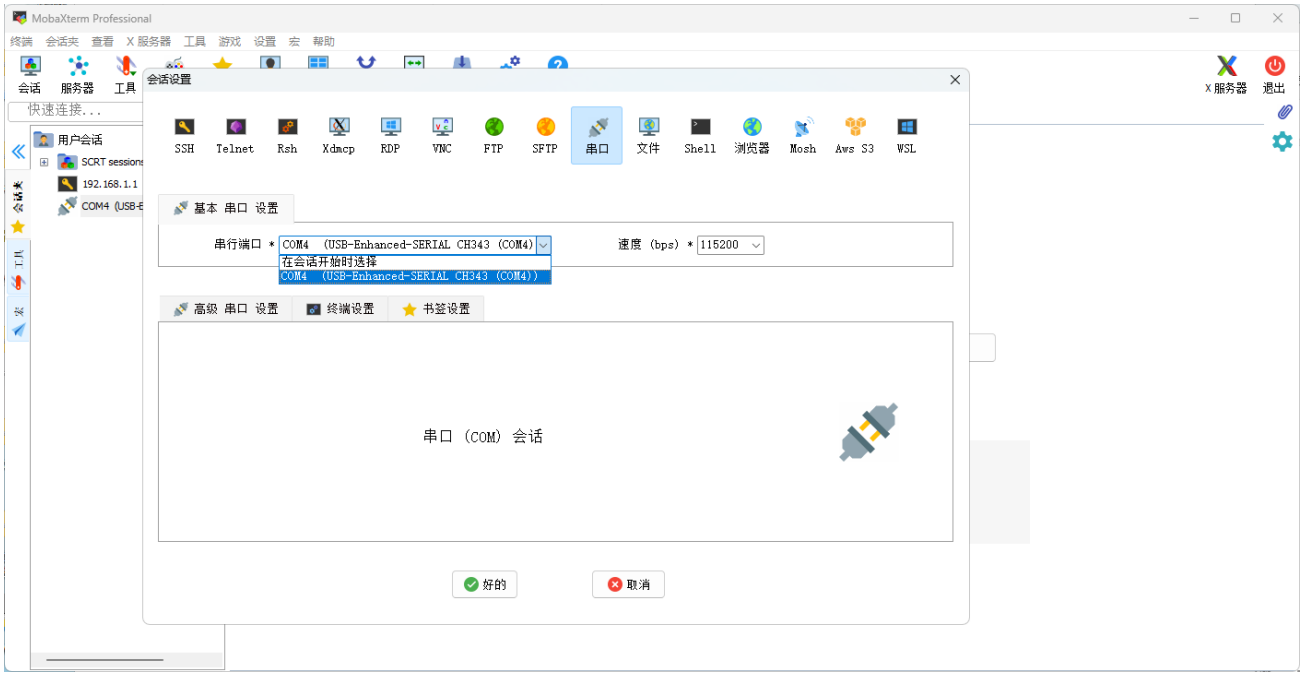


Figure 1-4 New serial session: pick the COM port, 115200 baud

1.5 Power on and interrupt autoboot

With the session open, power on the router. As soon as boot messages start scrolling, press space or enter rapidly until the ECNT> prompt appears — you are now at the U-Boot command line.

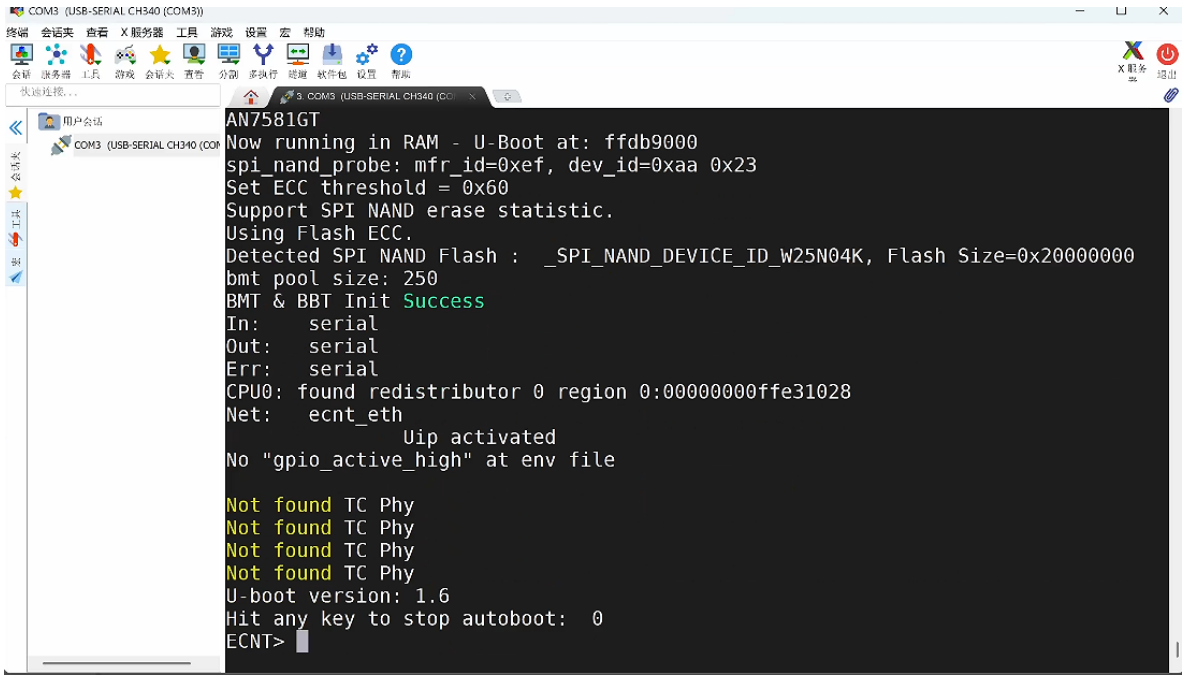


Figure 1-5 The ECNT> prompt means you are in the command line

> Missed the timing and the router booted normally? Just power-cycle and try again — it gets easier each time.

Chapter 2 Configure the PC network

Plug one end of the Ethernet cable into the router's yellow 1G port and the other end into the PC. Then set a fixed IP on the PC's wired adapter.

2.1 Set a manual IP address

Open Control Panel → Network and Internet → Network Connections, right-click Ethernet → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties, and fill in the values shown below:



Figure 2-1 Fill in the PC's IP address as shown

> Just copy the numbers from the screenshot. After flashing is complete, remember to switch back to Go back to "Obtain an IP address automatically" — otherwise the PC will not get online.

Chapter 3 Set up 3CDaemon, the transfer tool

3CDaemon is a small file server that runs on your PC. In chapter 4 the router downloads the bootloader file from it.

3.1 Put the bootloader file into the 3CDaemon folder

Copy the downloaded `xr1710g-chainloader-slot.bin` into the 3CDaemon folder (E:\3CDaemon\ in this example).



Figure 3-1 Bootloader file placed in the 3CDaemon folder

3.2 Confirm the upload/download directory

Open3CDaemon settingsfrom the left side of the main window and confirm the upload/download directory points to the same folder as the file (E:\3CDaemon\ here), then click OK.

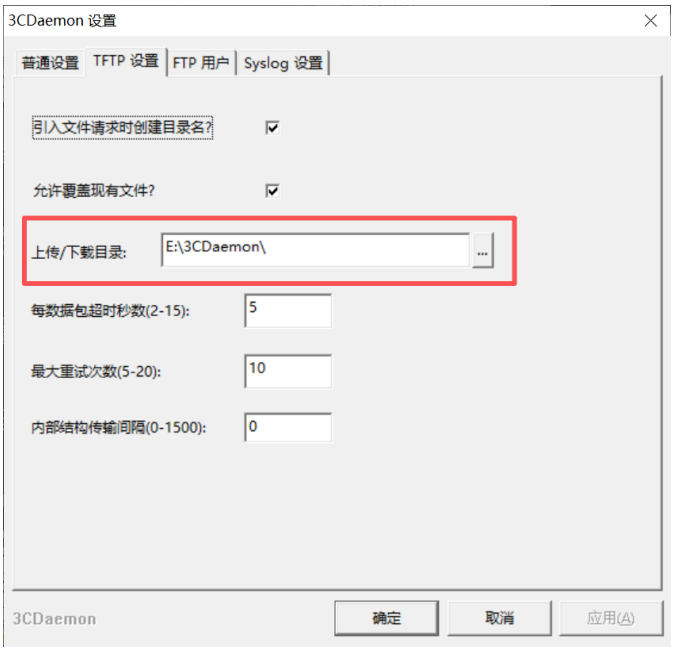


Figure 3-2 The upload/download directory must match the file location

3.3 Confirm the TFTP service is running

Back on the main window, the STOP badge on the left means the TFTP service is already running. Keep this window open.

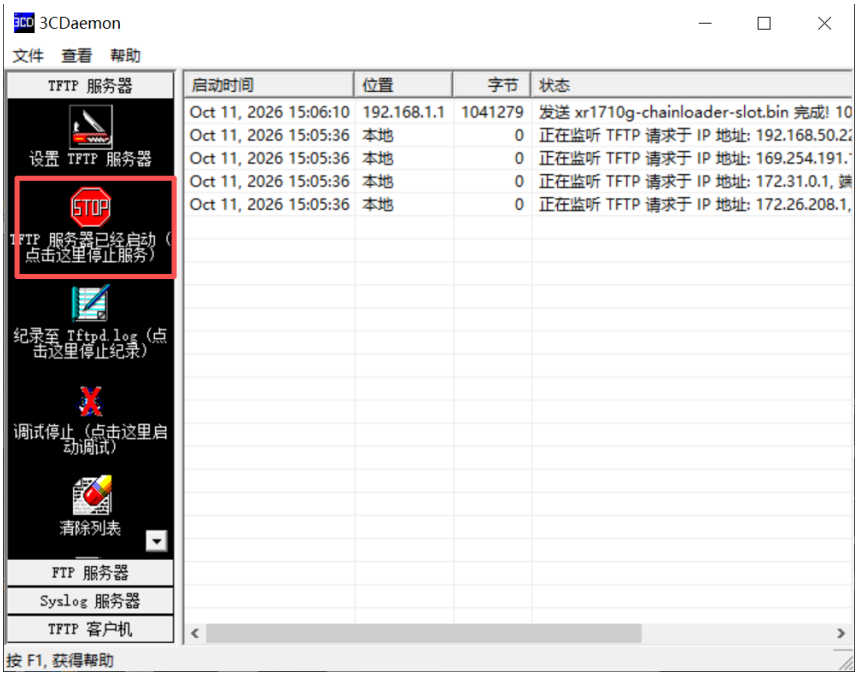


Figure 3-3 STOP on the left = the service is running

Chapter 4 Flash the U-Boot bootloader

Type the following 10 commands one line at a time at the ECNT> prompt, pressing enter after each line. Line 3 downloads the bootloader from your PC and takes a little while.

```
1 setenv serverip 192.168.255.10
2 setenv ipaddr 192.168.255.1
3 tftpboot 0x89000000 xr1710g-chainloader-slot.bin
4 flash erase 0x600000 0x100000
5 flash write 0x600000 0x100000 0x89000000
6 setenv one flash read 0x602100 0x0FDF00 \$loadaddr
7 setenv two "; bootm"
8 setenv bootcmd "$one$two"
9 saveenv
10 reset
```

4.1 If the transfer slows down or stalls

While line 3 runs, the Loading bar may fill with T characters (retransmission timeouts). That slows the transfer down but does **not** necessarily mean it failed. There is only one success criterion: **seeing done**. If the bar sits still for a long time without reaching done, retry like this:

1. Click STOP on the left of the 3CDaemon main window to stop the TFTP service;
2. Click it again to restart the service;
3. Back in the serial terminal, re-run line 3 (tftpboot) to transfer again.

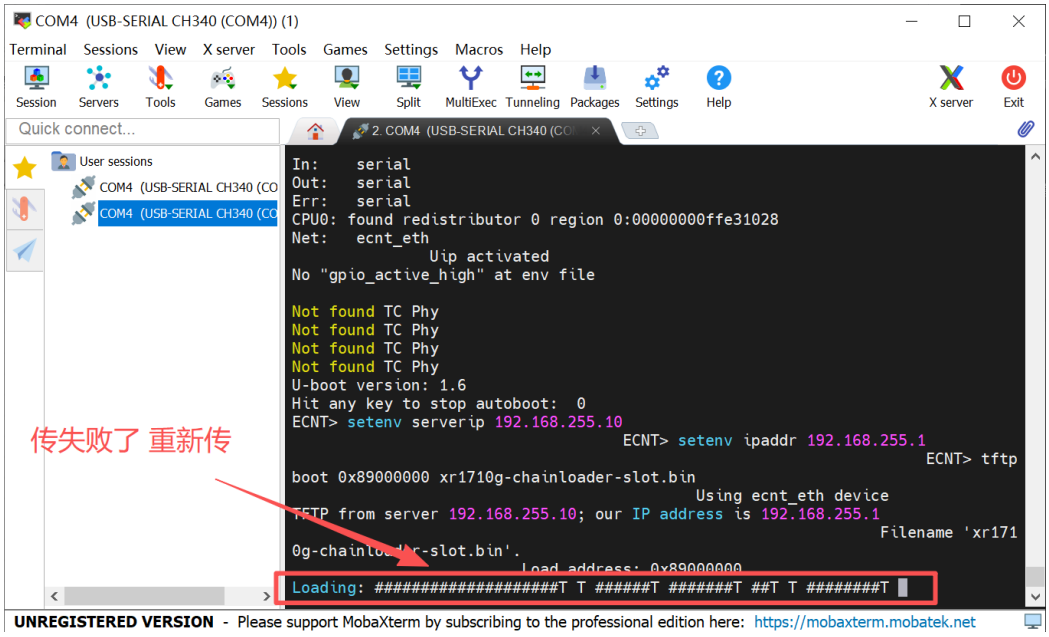


Figure 4-1 T characters in the bar: fine if done still appears; retry only when it stalls without done

4.2 The success marker: done

Once Loading finishes and you see done plus Bytes transferred = … (the number differs each run), the transfer succeeded — a few T characters along the way are perfectly fine. Then continue with lines 4–10.

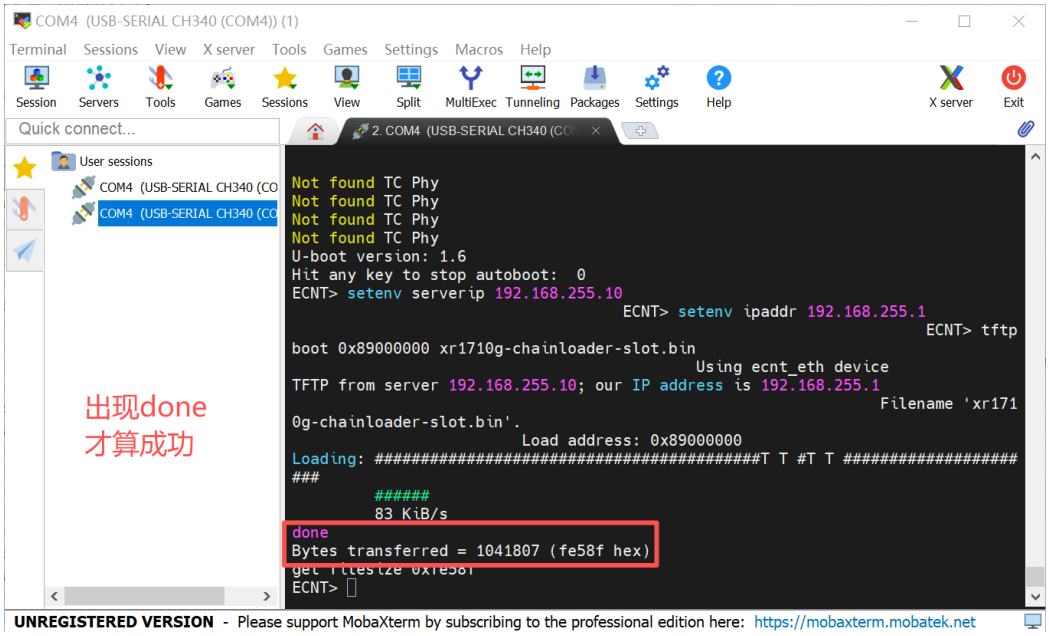


Figure 4-2 done means success (this run had T characters and still finished)

Chapter 5 Flash the Wiro iStoreOS firmware

5.1 Keep the network connected

Leave the Ethernet cable in the yellow 1G port and keep the fixed IP — 192.168.255.10 is on the same subnet as the router's 192.168.255.1.

5.2 Enter recovery mode

Power-cycle the router. Once the **red LED lights up**, press and hold the reset button (not sure about the timing? hold a few seconds longer — the margin is generous). When the status LED changes from solid red to a running marquee, recovery mode is active.

5.3 Open the Wiro recovery console

In a browser, open `192.168.255.1` to reach the Wiro recovery console'sSystem firmware double cleanpage.

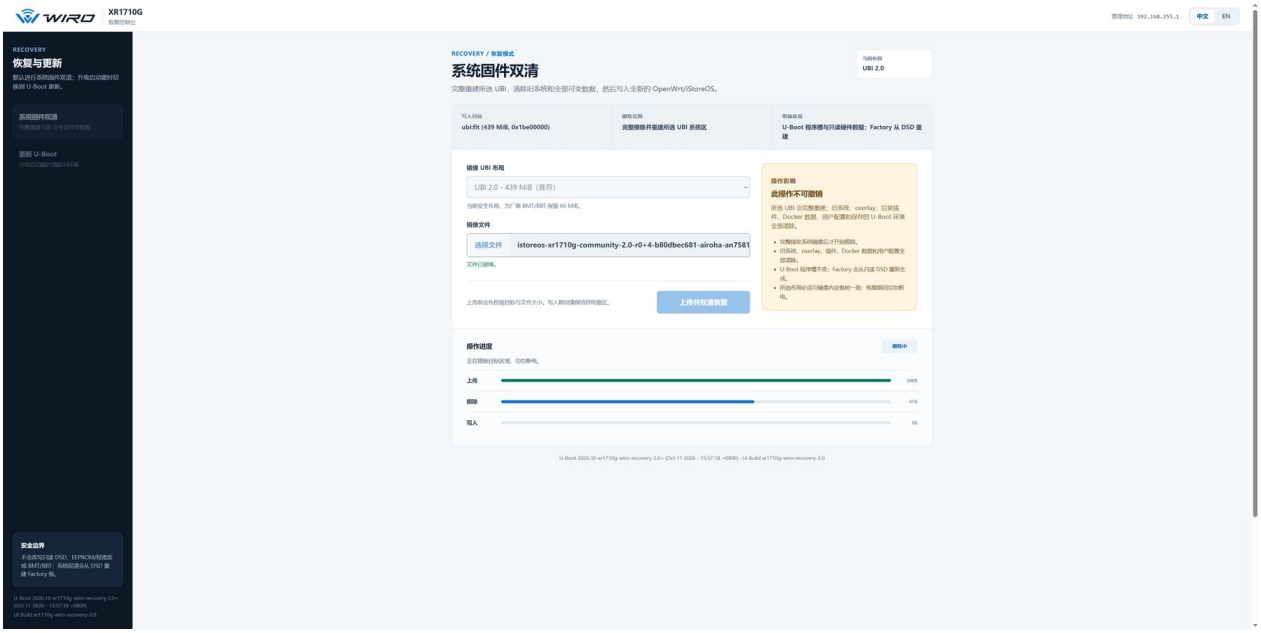


Figure 5-1 The Wiro recovery console: system firmware double clean

5.4 Upload the firmware and double clean

Click the file picker and select the downloaded istoreos-xr1710g-community firmware; keep the image UBI layout at its defaultUBI 2.0 (recommended); then click the upload-and-double-clean button.

> **The double clean wipes all configuration and data on the router and cannot be undone.** Never cut the power during upload, erase or write, or the device may no longer boot.

5.5 Wait for it to finish

The page shows three progress bars: upload → erase → write. When they all complete, the router reboots automatically and the firmware is installed.

Chapter 6 After flashing

6.1 Restore the PC network settings

Go back to the adapter properties from chapter 2 and switch the IP back to Switch back to "Obtain an IP address automatically", otherwise the PC may not get online.

6.2 Start using the new firmware

When the router has rebooted, follow the on-screen instructions of the Wiro iStoreOS firmware to join the Wi-Fi or open the administration page.

> Questions? **QQ group: 1061612207** · Paid remote flashing: **QQ 50848**.